Research Analysis Report

Title: Statistical Analysis of Risk Factors Associated with Coronary Heart Disease (CHD)
Prepared by: Chiemelie Onu
Role: Data Analysis Instructor
Date: October 2026

1.0 Executive Summary

This study examined factors associated with a 10-year coronary heart disease (CHD) outcome among 1,800 participants. The objective was to describe participant characteristics and identify factors independently associated with CHD. Descriptive statistics, cross-tabulations, chi-square tests, and multivariable binary logistic regression were used. The findings showed that age, sex, diabetes, and antihypertensive medication use were significantly associated with CHD, while education was not statistically significant and smoking showed a borderline association. Overall, the results highlight important demographic and cardiovascular risk-related factors associated with CHD in the study population.

2.0 Introduction

Coronary heart disease (CHD) is an important cardiovascular outcome associated with demographic, behavioural and clinical risk factors. Identifying factors associated with CHD can help improve understanding of population-level risk and inform preventive strategies.

2.1 Aim of the Analysis

Analytical objective

1. Describe the study population and cardiovascular risk-factor profile.
2. Investigate factors associated with the 10-year coronary heart disease outcome.

2.2 Research Questions & Objectives

2.2.1 Primary Objective

To identify demographic, clinical, and lifestyle factors independently associated with 10-year Coronary Heart Disease (CHD) risk within the study population.

2.2.2 Specific Objectives

I. To inspect, clean, and prepare the dataset for analysis while appropriately handling missing values and data quality concerns.
II. To perform univariable and bivariate analyses evaluating crude associations between candidate risk factors and the 10-year CHD outcome.
III. To construct an appropriate multivariable logistic regression model to estimate adjusted Odds Ratios (aOR) and 95% Confidence Intervals (CI) for independent predictors.
IV. To visually present key analytical findings and discuss potential data and model limitations.

2.2.3 Specific Research Questions

A. What are the demographic characteristics, baseline clinical profiles, and overall prevalence of 10-year CHD risk in the study population?
B. How does the occurrence of 10-year CHD vary across key participant subgroups and risk factors?
C. Which factors are independently associated with 10-year CHD after adjusting for potential confounders in a multivariable model?

3.0 Methods

1. Data Cleaning: Checked missing values, duplicates, coding, and data consistency.
2. Descriptive Analysis: Summarized categorical variables with frequencies/percentages and continuous variables with appropriate summary statistics.
3. Inferential Analysis: Used appropriate statistical tests to assess relationships and group differences.
4. Regression Analysis: Applied logistic regression to identify factors associated with the binary outcome.
5. Statistical Significance: Reported effect estimates, 95% confidence intervals, and p-values, using $p < 0.05$ as the significance level.
6. Software: Analysis was conducted using IBM SPSS Statistics, with syntax retained for reproducibility.

3.1 Data Overview, Quality Checks and Preparation

The dataset contains 1,800 participant cases and cardiovascular health variables covering demographics, lifestyle factors, clinical measurements, cardiovascular risk factors, and 10-year CHD outcome. Variables include age, sex, education, smoking, diabetes, blood pressure, cholesterol, BMI, heart rate, glucose, and CHD status.

1. Variable labels were added to provide clear descriptions of the variables and their respective meanings.

2. Value labels were assigned to categorical variables where applicable to ensure clear interpretation of coded responses.

3. The missing value for level of education attained was coded as −999 to explicitly identify it as a missing observation.

4. Missing values for average number of cigarettes smoked per day were coded as −999.

5. Missing values for use of antihypertensive medication at examination were coded as −999.

6. The missing value for body mass index (BMI) was imputed using the mean BMI value of the available observations.

7. A new age-group variable was created to facilitate analysis of participants by age categories.

3.2 Statistical Analysis

Occurrence of the 10-Year CHD Outcome

Among 1,800 participants, 273 (15.2%) experienced a 10-year CHD outcome. CHD occurrence was higher among males than females (20.1% vs. 11.5%) and increased with age, from 5.6% in those aged 33–44 years to 37.5% in those aged 65–69 years. It was also higher among participants with hypertension (24.6%) and diabetes (42.9%) than those without these conditions (10.9% and 14.5%, respectively). CHD occurrence was similar among current smokers and non-smokers (15.6% vs. 14.7%). Overall, CHD was more common among older participants, males, and those with hypertension or diabetes.

Table 1. *Participant Characteristics and 10-Year CHD Outcome*

Variable	**N**	**Summary**
Age (years)	1800	49.73 ± 8.54
Total cholesterol (mg/dL)	1780	236.10 ± 44.23
Systolic BP (mmHg)	1800	132 .44 ± 22.28
Diastolic BP (mmHg)	1800	82.92 ± 12.14
BMI (kg/m²)	1794	25.92 ± 4.15
Current smoker		
No	898	49.90%
Yes	902	50.1%
Diabetes		
No	1758	97.7%
Yes	42	2.3%
Hypertension		
No	1234	68.6%
Yes	566	31.4%
10-year CHD outcome		
No	1527	84.8%
Yes	273	15.2%

Note*: Continuous variables are presented as mean ± SD and categorical variables as n (%).*

Table 2. *Multivariable Logistic Regression for 10-Year CHD Outcome*

Predictor	Comparison	OR	*p*-value
Age	Per 1-year increase	1.085	<.001
Sex	Female vs male	0.530	<.001
Education	Overall		.416
Smoking	Non-current vs current	0.748	.051
Diabetes	Yes vs no	3.09	.001
Antihypertensive Medication	Current use vs non-use	2.32	.006

Model statistics: $\chi^2(8) = 157.47$, $p < .001$; Nagelkerke $R^2 = 0.152$; N = 1,726.

3.3 Interpretation of Main Effect Estimates

A multivariable binary logistic regression model examined factors independently associated with 10-year CHD outcome. The overall model was statistically significant, $\chi^2(8) = 157.467$, $p < .001$.

Age was significantly associated with CHD; each additional year was associated with 8.5% higher odds of CHD (OR = 1.085, $p < .001$).

Sex was also significantly associated with CHD. Females had approximately 47% lower odds of CHD than males (OR = 0.530, $p < .001$), after adjusting for other predictors.

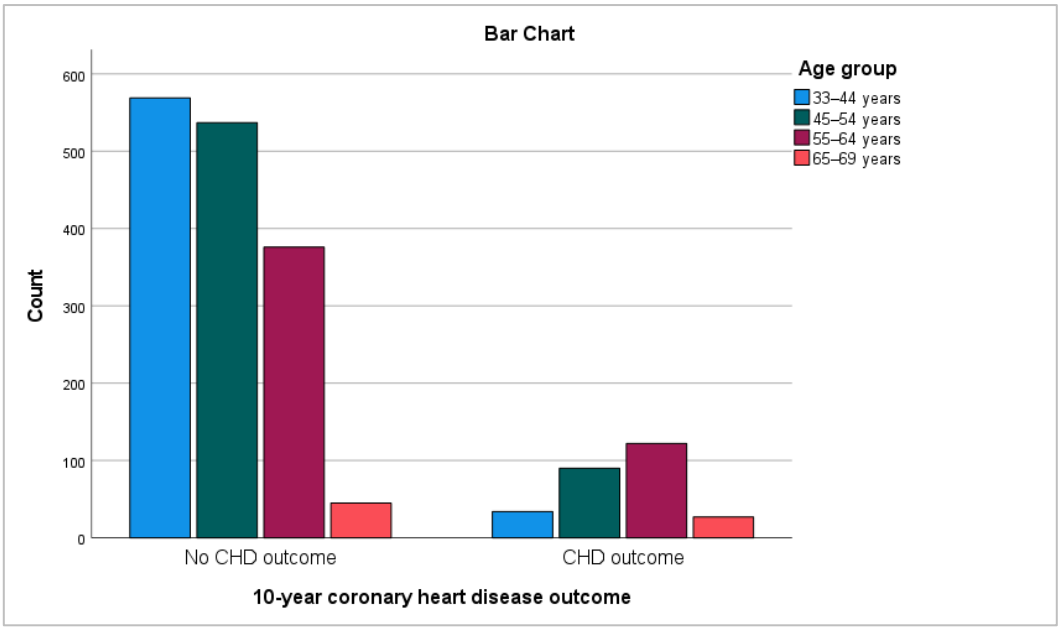
Education level was not significantly associated with CHD ($p = .416$). Smoking status was borderline but did not reach statistical significance ($p = .051$).

Diabetes was significantly associated with CHD. Participants with diabetes had approximately 3.09 times the odds of CHD compared with those without diabetes (OR = 3.09, $p = .001$).

Antihypertensive medication use was also significantly associated with CHD. Current users had approximately 2.32 times the odds of CHD compared with non-users (OR = 2.32, $p = .006$). This association should be interpreted cautiously, as medication use may reflect underlying cardiovascular risk.

Overall, age, sex, diabetes, and antihypertensive medication use showed significant independent associations with CHD, while education was not significant and smoking was borderline.

Figure 1. *Distribution of 10-year CHD outcome across age groups.*



3.4 Key Findings and Limitations

1. Age was strongly associated with CHD. Each additional year of age was associated with an 8.5% increase in the odds of CHD (OR = 1.085, $p < .001$).

2. Diabetes and antihypertensive medication use were associated with higher CHD odds. Participants with diabetes had approximately 3.09 times the odds of CHD (OR = 3.09, $p = .001$), while current medication users had approximately 2.32 times the odds (OR = 2.32, $p = .006$).

3. Sex was significantly associated with CHD. Females had approximately 47% lower odds of CHD than males (OR = 0.530, $p < .001$). Education was not significant, while smoking was borderline ($p = .051$).

3.5 Limitations and interpretation issues

The analysis used complete cases, excluding participants with missing values from the regression model, which may introduce bias if missingness was not random. The model also showed low sensitivity for identifying CHD cases, so the overall classification accuracy should not be interpreted as strong predictive performance. In addition, the observed associations are not evidence of causation. The association between antihypertensive medication use and CHD, for example, may partly reflect underlying cardiovascular risk or hypertension rather than an effect of the medication itself.

Appendix A

Variable Dictionary/Coding

Variable	Description	Type	Coding / Units
participant_id	DUAC-generated assessment record identifier	Identifier	CVR0001, CVR0002, ...
sex_male	Participant sex indicator	Binary	1 = Male; 0 = Female
age_years	Age at examination	Numeric	Years
education_level	Attained education level	Ordinal categorical	1 = 0–11 years; 2 = High school diploma/GED; 3 = Some college/vocational school; 4 = College degree or more
current_smoker	Current cigarette smoking status	Binary	1 = Current smoker; 0 = Not a current smoker
cigarettes_per_day	Average number of cigarettes smoked per day	Numeric	Cigarettes/day; 0 for non-smokers
bp_medication	Use of antihypertensive medication at examination	Binary	1 = Current use; 0 = Not currently used
prevalent_stroke	Indicator of prevalent/prior stroke	Binary	1 = Yes; 0 = No
prevalent_hypertension	Indicator of prevalent hypertension	Binary	1 = Yes; 0 = No
diabetes	Diabetes indicator	Binary	1 = Yes; 0 = No
total_cholesterol_mg_dl	Serum total cholesterol	Numeric	mg/dL
systolic_bp_mmhg	Systolic blood pressure	Numeric	mmHg
diastolic_bp_mmhg	Diastolic blood pressure	Numeric	mmHg
bmi_kg_m2	Body mass index	Numeric	kg/m²
heart_rate_bpm	Heart rate	Numeric	Beats per minute
glucose_mg_dl	Casual serum glucose	Numeric	mg/dL
ten_year_chd	Binary 10-year coronary heart disease outcome/target supplied in the teaching extract	Binary outcome	1 = CHD outcome; 0 = No CHD outcome